12th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering 2026 (IEEE WIECON-ECE 2026)

Recalibration Data Requirements for Battery Failure Prediction Under Operational Distribution Shift

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Abstract—Probabilistic battery failure prediction models are trained on clean laboratory cycling data and deployed under operating conditions that differ from the training distribution. This paper asks how much field data is needed to restore the calibration of a frozen multihorizon hazard model after such a shift. Partial discharge, temperature noise, and rest irregularity are applied at four severity levels with five seeds each. Expected calibration error (ECE) rises from 0.23–0.41 on clean held-out data to 0.31–0.51 under perturbation and saturates at the mildest severity. Refitting the isotonic map on a field sample cuts perturbed ECE monotonically, from a mean of 0.221 at a 5% sample to 0.121 at 20%, with only marginal gains beyond. The 20% requirement holds at every horizon. Domain-randomized retraining does not beat this lightweight recalibration at any horizon except H = 10. The practical recommendation is to keep the base model and refit its calibrator on about one-fifth of a field window.

Keywords—battery energy storage, calibration, distribution shift, hazard learning, isotonic regression, recalibration

I. Introduction

Battery energy storage systems are dispatched for grid services on a routine basis. Operators need more than the energy a battery can deliver. They need a trustworthy probability that the battery completes a committed service within a predefined horizon, because dispatch decisions are gated by probability thresholds. A multihorizon hazard model predicts, for each cycle, the probability of failure within 10 to 50 cycles and uses isotonic regression to turn raw scores into calibrated probabilities. When the probabilities drift, threshold-gated dispatch makes unsafe or uneconomical decisions.

Laboratory-trained models meet three systematic shifts in the field: partial cycles at varying depth of discharge (DoD), irregular rest intervals, and fluctuating thermal conditions. The prognostics literature has mapped state-of-health (SOH) estimation and remaining-useful-life (RUL) prediction in detail, and the calibration-under-shift literature has grown in image and NLP domains. The operational question for batteries has not been answered: when calibration degrades, how much field data restores it?

This paper answers that question with a controlled stress test of a frozen hazard model under the three shifts just described. Three contributions follow. First, we quantify how much calibration degrades under synthetic operational perturbations, spanning four prediction horizons and four severity levels from mild to aggressive. Second, a recalibration data-requirement curve shows the field-sample fraction needed to restore calibration. Third, recalibration is compared against domain-randomized retraining, which offers no consistent advantage over the light recalibration step. For operators this translates to a deployment recipe: refit the calibrator on about a fifth of a field window, keeping the base model frozen.

II. Related Work

A. Battery health prognostics

Battery health prognostics has concentrated on SOH estimation and RUL prediction. Surveys by Che et al. [1] and Wang et al. [2] catalogue data-driven approaches. Representative methods include deep learning for SOH without degradation experiments [3], optimized XGBoost for capacity prediction [4], and cycle-based deep architectures [5]; Massaoudi et al. [6] review deep-learning prognostics. Hybrid and EIS-based RUL models [7]–[9], joint SOH/RUL prediction from single-cycle or discharge-fragment data [10], [11], and transfer learning for fast-charging profiles [12] round out the recent work. A common limitation is that validation is confined to clean laboratory splits of a single dataset. Performance under operational distribution shift is not characterized.

B. Probability calibration

Probability calibration has a long lineage. Platt [13] introduced sigmoid scaling for support vector machines; Zadrozny and Elkan [14] extended calibration to tree ensembles; Niculescu-Mizil and Caruana [15] showed that boosted ensembles exhibit characteristic sigmoidal distortion. Naeini et al. [16] formalized ECE. Guo et al. [17] demonstrated that modern models are poorly calibrated in-distribution and that temperature scaling helps. Temperature scaling remains the default post-hoc method because it fits a single parameter, but that parametric form is too restrictive when the distortion is nonlinear, which is the regime produced by operational shift. Isotonic regression, used here, is the nonparametric counterpart, at the price of a fitted sample.

C. Calibration under distribution shift

Ovadia et al. [18] benchmarked predictive uncertainty under dataset shift and found calibration collapses before discrimination. Levi et al. [19] extended calibration evaluation to regression. Gawlikowski et al. [20] consolidated the uncertainty literature; Angelopoulos and Bates [21], Zhang and Wang [22], and Gibbs and Candès [23] provided conformal alternatives with distribution-free guarantees; Wu [24] surveyed emerging calibration and shift problems. This literature is dominated by image and NLP benchmarks. Batteries bring a specific constraint: the shift is structured (partial discharge truncates the voltage curve directly), and the open deployment question is the cost of restoring calibration once it degrades.

D. Domain randomization

Domain randomization trains on perturbed copies of the source data so the model learns shift-invariant representations. It has succeeded in sim-to-real robotics [25], [26]. Its transfer to battery hazard models is untested, and retraining a hazard model on augmented data is not obviously cheaper than recalibrating the output. This paper evaluates both. Domain randomization assumes the augmentation family spans the deployment distribution; when the mismatch is structural rather than statistical, the learned invariance is only partial, and the cost scales with a full retraining.

E. Gap

No prior study reports how much field data recalibration requires for a battery hazard model under operational shift. That gap is closed here with an explicit sample-fraction sweep and paired bootstrap confidence intervals.

III. Background

A. Hazard prediction and the composite failure label

Operational failure is defined as the inability of a cell to complete a service commitment within a horizon H. A composite label marks failure at the earliest of two events: a voltage sag below 94% of the early-cycle baseline (averaged over the first 10 cycles of the cell) or an SOH drop below 80%. Let V_min,t and SOH_t denote the per-cycle minimum voltage and state of health, and V̄_base the early-cycle baseline. The failure label is

$$y_t = 1 \;\; if \;\; \left(V_{min,t} \leq 0.94 \cdot \overline{V}_{base}\right) \vee \left(SOH_t \leq 0.80\right),\; 0 \;\; othe \qquad (1)$$

Every cycle with index t satisfying t + H ≥ t_fail is then marked pre-failure for that horizon:

$$y_t^H = 1 \;\; if \;\; t + H \geq t_{fail},\; 0 \;\; otherwise \qquad (2)$$

A gradient-boosted tree ensemble maps the per-cycle feature vector x_t = {SOH_t, V_avg,t, V_min,t, I_avg,t, T_avg,t, duration_t, t} to a failure probability fθ(x_t). The ensemble is trained on clean data, and an isotonic map is fitted on out-of-fold predictions. Both the classifier and the calibrator are then frozen. Two properties of the label shape the analysis below. First, the failure label is compositional: a cell can break its service commitment (voltage sag) or its health commitment (SOH), and the earliest event triggers the pre-failure window. The 94% threshold is computed per cell against its own early-cycle baseline, so the label keeps a fixed semantic meaning across cells. Second, the pre-failure label is not a Markov state: once a cell crosses t_fail, every subsequent cycle in the window is labeled, so consecutive cycles are strongly correlated. The horizon H therefore sets both the label density and the operational lead time.

B. Calibration error

A model is calibrated when P(failure | P̂ = p) ≈ p for all p. ECE approximates the gap between predicted probability and observed frequency with B bins, where n_b is the bin population, N the sample size, acc_b the observed frequency, and p̄_b the mean predicted probability:

$$ECE = \sum_{b=1}^{B} \frac{n_b}{N} \cdot \left| acc_b - \overline{p}_b \right| \qquad (3)$$

ECE is computed here with ten equal-width bins, the last bin closed. It is the metric used throughout this paper because dispatch consumes probability magnitudes rather than rank order. Ten bins balances resolution against sample size: with 163 evaluation records, finer binning leaves several bins with fewer than a dozen samples and inflates variance.

IV. Experimental Design

A. Dataset

Experiments use the NASA Ames lithium-ion battery dataset [27]: 37 LCO 18650 cells cycled to end of life under varying conditions, yielding 1,028 valid cycles after SOH filtering. Cycle-life models for other chemistries follow different degradation dynamics [28]. Cells are split by physical battery ID with an 80/20 holdout, so no cell seen by the calibrator appears in the evaluation set (7 test batteries, 163 held-out cycles). The dataset ages cells under a range of charge regimes and chamber temperatures (24–43 °C), so the held-out partition is not a replication of the training cells' conditions; the clean baseline already contains a mild, uncontrolled cross-cell shift, and the controlled perturbations of Section IV-C are applied on top of it.

B. Model training

An XGBoost classifier [29] is trained on the clean training partition (Table I). The isotonic calibrator is fitted on out-of-fold predictions from 5-fold GroupKFold within the training partition. The result is a pure observational-shift test: perturbations affect only the input distribution, never the model.

TABLE I. MODEL HYPERPARAMETERS (TRAINING CONFIGURATION)

Parameter	Value
Model type	XGBoost
Estimators / depth / learning rate	300 / 4 / 0.05
Subsample / colsample	0.8 / 0.8
Min child weight	5
Calibration	Isotonic regression (OOF fit)

C. Perturbation model

Raw discharge curves are transformed into operational profiles by three mechanisms whose intensity increases across four severity levels (Table II). Partial cycling truncates each discharge curve at a randomly sampled DoD fraction, with the sampling range tightening at higher severity. The truncated duration of cycle t is

$$d_t \sim U(l_s, u_s), T_t' = (1 - d_t) \cdot T_t \tag{4}$$

All seven features are recomputed from the truncated signal, so duration falls while average and minimum voltage rise as the discharge tail is cut. Temperature noise is added to raw temperature measurements before averaging:

$$\bar{T}_{avg}' = \frac{1}{m} \sum_{i=1}^{m} (T_i + \varepsilon_i) \tag{5}$$

Rest irregularity applies proportional Gaussian noise to duration and average temperature, so each feature x_t is scaled as

$$x_t' = x_t \cdot (1 + \eta_t), \eta_t \sim N(0, \rho_s^2) \tag{6}$$

The true SOH trajectory and failure labels are preserved, isolating observational shift from changed degradation dynamics. Each severity is realized with five seeds (42, 123, 456, 789, 101112), producing 20 perturbed datasets and 20,560 records in total.

TABLE II. PERTURBATION SEVERITY LEVELS (APPLIED TO RAW CURVES)

Level	DoD truncation	σs (°C)	ρs
S1 mild	α=0.15 (75–100% DoD)	0.5	0.01
S2 moderate	α=0.30 (55–75% DoD)	1.0	0.02
S3 severe	α=0.50 (35–55% DoD)	2.0	0.03
S4 aggressive	α=0.85 (15–35% DoD)	3.0	0.05

Because truncation operates on the raw curve, the shift is physically coupled: shorter discharges lower duration and raise average and minimum voltage together, while temperature statistics are perturbed independently. A feature-level augmentation would miss this coupling, which is why the perturbation is applied before feature extraction rather than on the feature vectors themselves.

D. Evaluation protocol

For every (severity, seed, horizon) condition, ECE is computed for the frozen model and for a recalibrated model whose isotonic map is refitted on an independent field sample of the perturbed data. The recalibration map minimizes squared error under a monotonicity constraint over the calibration set D_cal:

$$\widehat{m} = \arg\min_{m\ monotone} \sum_{i \in D\_cal} (y_i - m(p_i))^2 \tag{7}$$

The field-sample fraction is swept over 5%, 10%, 20%, and 50%, sampled with a dedicated random number generator. All metrics are reported on the same held-out subset, so paired comparisons are never confounded by differing evaluation samples. Gains are summarized with paired bootstrap 95% confidence intervals (10,000 resamples per horizon). Because every metric is evaluated on the same records, the bootstrap pairs are matched within each (severity, seed, horizon) condition, so the intervals in Table VI quantify variability across conditions rather than label noise.

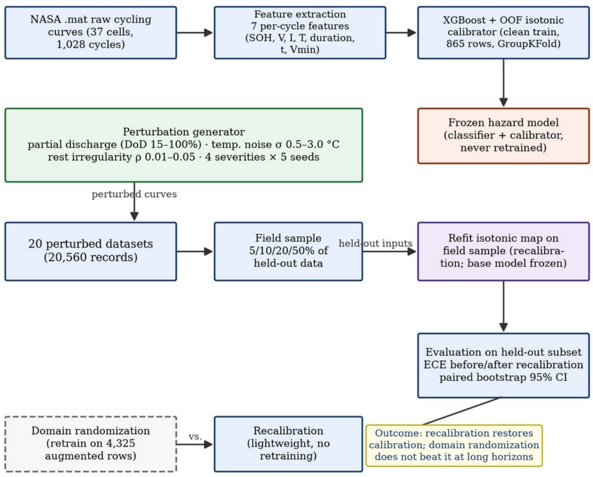

Fig. 1. Methodology: a frozen XGBoost hazard model with out-of-fold isotonic calibration is stress-tested under synthetic operational perturbations; recalibration on a field sample is compared with domain-randomized retraining.

V. Results

A. Clean baseline

On the strictly held-out test partition, the frozen model's calibration is weak even on clean data: ECE ranges from 0.231 at H = 20 to 0.413 at H = 50 (Table III). Because that baseline is already weak, the degradation we report below is conservative. Discrimination is modest throughout (AUC ≈ 0.48–0.59), which is typical of autocorrelated pre-failure labels. The practical headroom sits in probability magnitudes, and horizon shifts calibration even without any perturbation.

TABLE III. CLEAN BASELINE ON THE HELD-OUT TEST PARTITION

H	ECE raw	ECE cal	AUC cal	Brier
H=10	0.375	0.337	0.578	0.253
H=20	0.336	0.231	0.594	0.213
H=30	0.380	0.369	0.593	0.332
H=50	0.492	0.413	0.485	0.410

B. *Calibration degrades and saturates*

Under perturbation, ECE rises at three of four horizons, from 0.231 to 0.311 at H = 20 and from 0.413 to 0.507 at H = 50, with little change at H = 10 (0.337 to 0.328; Table IV). The degradation saturates at the mildest severity: mean perturbed ECE is flat across severity levels, at 0.395, 0.384, 0.372, and 0.382 for S1 through S4 (Fig. 3). Fig. 4 explains the saturation. In full discharge Vmin reaches about 2.3 V; at the mildest truncation it jumps to about 3.2 V, a 37% shift the frozen calibrator cannot absorb. The reliability diagrams at H = 20 (Fig. 2) show the same pattern: every severity curve bends below the diagonal in the same probability region, and the curves cluster instead of spreading, confirming that saturation is a property of the shift rather than of severity. Within each severity the shift is dominated by the truncation draw rather than the noise level: mean ECE varies by at most 0.047 between S1 and S4 at any horizon, while the clean-to-perturbed gap reaches 0.094 at H = 50. Discrimination is stable by contrast: AUC moves by less than 0.1 at every horizon (0.594 → 0.609 at H = 20) and does not track the ECE collapse. The perturbation shifts probability magnitudes while ranking quality stays intact.

TABLE IV. CLEAN AND PERTURBED ECE ON THE HELD-OUT PARTITION (MEAN ACROSS 4 SEVERITIES × 5 SEEDS)

H	ECE clean	ECE perturbed
H=10	0.337	0.328
H=20	0.231	0.311
H=30	0.369	0.388
H=50	0.413	0.507

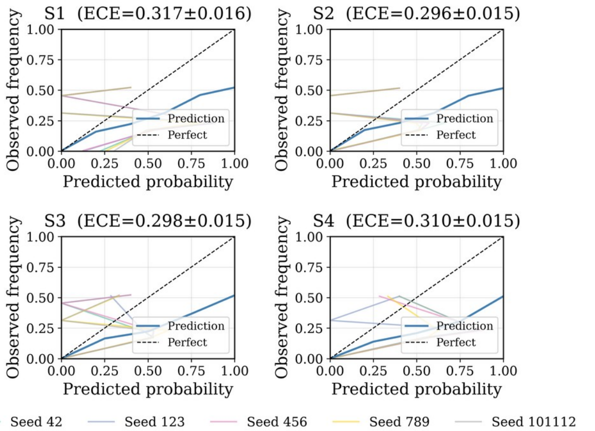

Fig. 2. Reliability diagrams at H = 20 across severities S1–S4 (five seeds each). Deviation from the diagonal saturates at the mildest perturbation.

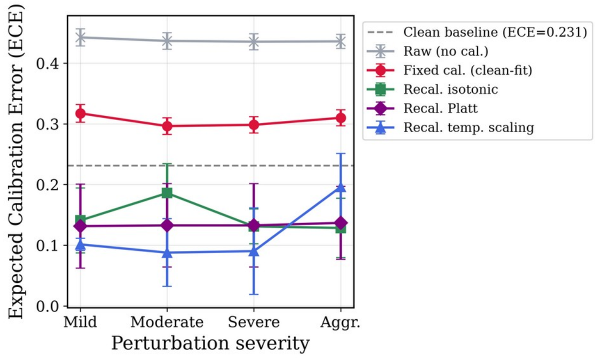

Fig. 3. ECE vs. severity level for each horizon. Degradation does not grow with severity; the mildest level already carries most of the shift.

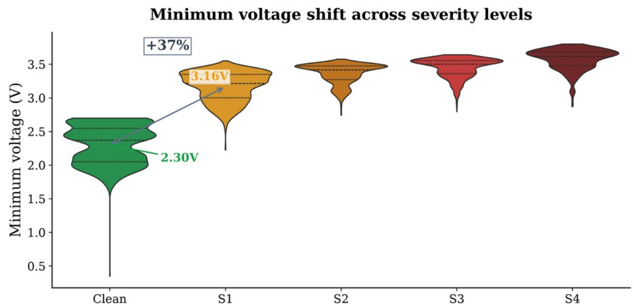

Fig. 4. Minimum-voltage distribution, clean vs. perturbed (seed 42). The mean shifts from 2.3 V to 3.2 V by severity 1, driving the calibration collapse.

C. *Recalibration data requirement*

Refitting the isotonic map on an independent field sample restores calibration, and the requirement is small. Mean held-out ECE falls monotonically with sample fraction, from 0.221 at 5% to 0.175 at 10%, 0.121 at 20%, and 0.080 at 50% (Fig. 5, Table V). The improvement saturates near a 20% fraction at every horizon. Paired bootstrap intervals for the 10% configuration exclude zero at all horizons, with gains from 0.141 [0.113, 0.169] at H = 10 to 0.358 [0.333, 0.382] at H = 50 and Cohen's d_z from 1.8 to 6.2 (Table VI). For a deployment window of a few hundred cycles, a 20% sample is a small collection effort, and the base model stays untouched. The curve is steep between 5% and 20% and flattens beyond: the marginal gain from 20% to 50% is only 0.041 in mean ECE, which bounds the operator's collection effort. The requirement is also roughly the same at every horizon: at a 20% fraction, per-horizon means cluster between 0.098 (H = 10) and 0.149 (H = 30), within 0.05 of each other. Seed-to-seed spread of ECE at a 10–20% fraction is 0.06–0.08, well below the gains of Table VI, so the requirement is stable across individual augmentations. The paired rank-biserial correlation between recalibrated and uncalibrated ECE is 1.0 at every horizon: every condition improves, and no single condition drives the mean.

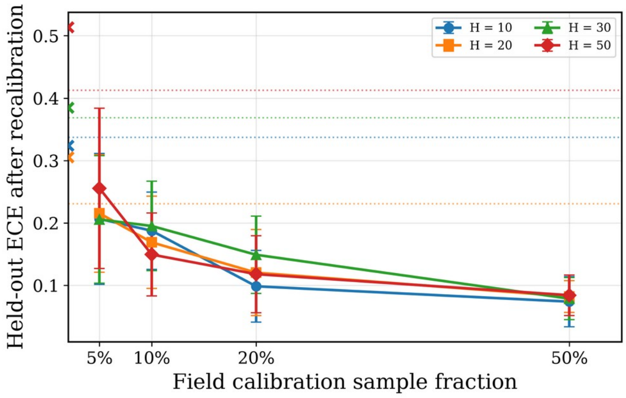

Fig. 5. Held-out ECE after recalibration vs. field sample fraction, per horizon (mean ± std across severities and seeds). Markers at 2% show perturbed ECE without recalibration; dotted lines show the clean baseline per horizon.

TABLE V. RECALIBRATED ECE VS. FIELD SAMPLE FRACTION (MEAN ACROSS 4 SEVERITIES × 5 SEEDS)

Fraction	H=10	H=20	H=30	H=50	Mean
5%	0.206	0.215	0.206	0.255	0.221
10%	0.187	0.169	0.195	0.149	0.175
20%	0.098	0.120	0.149	0.118	0.121
50%	0.074	0.082	0.079	0.084	0.080

TABLE VI. RECALIBRATION GAIN AT A 10% FIELD SAMPLE (PAIRED BOOTSTRAP, 20 PAIRS PER HORIZON)

H	ECE gain	95% CI	Cohen's d_z
H=10	0.141	[0.113, 0.169]	2.18
H=20	0.142	[0.107, 0.176]	1.76
H=30	0.193	[0.162, 0.223]	2.75
H=50	0.358	[0.333, 0.382]	6.21

D. *Domain randomization is not competitive*

Retraining the base classifier on perturbed data is the standard alternative to post-hoc recalibration. A fresh XGBoost is trained on the clean training set augmented with perturbed copies at all four severities, for a total of 4,325 rows (the 865-row clean set plus four perturbed copies). Domain randomization wins ECE only at H = 10, at 0.074 against 0.187, and loses at every longer horizon, where recalibration at a 10% sample reaches 0.149–0.195 while domain randomization stays between 0.314 and 0.381. The added complexity of generating perturbed training data and retraining is not justified. Lightweight recalibration remains the preferred deployment strategy. The gap widens with horizon: at H = 50 domain randomization reaches 0.381 against 0.149 for recalibration.

VI. Discussion

A. *An operational boundary map*

Aggregating across conditions yields three operating zones. Direct deployment of the frozen model is unsafe under partial cycling: ECE exceeds 0.28 at most (horizon, severity) pairs. Recalibrated operation enters a warning zone (ECE 0.10–0.20) at all horizons and reaches the safe zone (ECE < 0.10) at several horizons. Recalibration is therefore not optional for field deployment; the open question is only how quickly it can be performed, and Fig. 5 shows the answer is about one-fifth of a field window. The map also doubles as a monitoring instrument: ECE can be re-estimated on each recalibration sample, so a drift back toward the warning zone flags that a refresh is due before reliability degrades.

B. *Deployment recipe*

The results support a concrete procedure: deploy the frozen model, collect about 20% of a field window from normal operation, and refit a fresh isotonic map on the raw probabilities versus observed outcomes using the same seven per-cycle features (Fig. 6). Redeploy with the recalibrated map, and refresh the calibrator periodically as conditions drift. None of the steps touches the base model, the feature set, or the original training data. The recipe assumes the shift is static within a refresh window; under continuous drift the interval must shrink, and the monitoring loop of Section VI-A flags when it should.

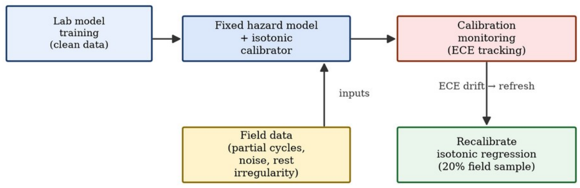

Fig. 6. Calibration-aware deployment: the frozen hazard model is paired with a lightweight recalibration step on field data.

C. *Limitations*

Several caveats bound the conclusions. The perturbation model preserves the true SOH trajectory, isolating observational shift but not the coupled degradation dynamics of real partial cycling. Results are for a single LCO chemistry, a single model family, and a single 163-record held-out partition. Temperature noise is Gaussian and does not model thermal gradients or diurnal variation. Replication on larger multi-chemistry datasets is warranted, and the ECE-only metric could be extended with MCE or ACE. Operationally, recalibration assumes outcome labels for the field sample; in a deployed fleet these require an on-line SOH estimate, itself a model output, so the loop inherits that estimator's noise. Threshold monitoring (Section VI-A) mitigates this by triggering a refresh before drift accumulates. None of these caveats changes the practical conclusion: calibration of a frozen battery hazard model degrades sharply under operational shift, and refitting the calibrator on roughly 20% of a deployment window restores it.

VII. Conclusion

This paper quantified the cost of restoring calibration for a frozen battery hazard model that must operate under operational distribution shift. Three findings stand out. Calibration is fragile: ECE rises from 0.23–0.41 on clean data to 0.31–0.51 under perturbation and saturates at the mildest severity. The damage is cheap to repair. A 20% field sample suffices to refit the isotonic map at every horizon, with gains well separated from zero. Domain-randomized retraining, the conventional alternative, never beats this lightweight maintenance step at longer horizons. Operators can keep their validated base model. Periodic recalibration on a modest field sample maintains trustworthy probabilities. Future work targets non-Gaussian and coupled perturbation families that move degradation dynamics as well as observations, adaptive refresh schedules triggered by stream ECE, and replication across chemistries and model families.

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